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Deuterium hybrid X-pinch driven by small dense plasma focus

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Abstract

This study presents a proof of the concept of a deuterium hybrid X-pinch configuration, driven by a 3 kJ dense plasma focus (PFZ-200) at the Czech Technical University in Prague. The system consists of two opposing conical electrodes filled with deuterium gas, forming a point-like source of DD fusion protons via D(d,p)T reactions. Two axial anode–cathode gaps of 3 and 5 mm were investigated. Key diagnostics included CR-39-based proton imaging, pinhole camera, and energy spectrometry. The smaller A–K gap (3 mm) produced a compact, nearly circular proton source with a diameter of 1.2 mm and yields on the order of 10^7 protons/shot. Based on the spectral measurements, the maximum proton energy reached 3.6 MeV, corresponding to deuteron energies approximately up to 1.3 MeV. Compared to standard plasma focus operation, the hybrid X-pinch configuration provided improved proton source localization suitable for advanced plasma diagnostics such as proton imaging and deflectometry.

1. Introduction

Z-pinch discharges are known to be intense x-ray, neutron and ion sources. Many configurations such as fiber or wire array z-pinch, gas-puff z-pinch, X-pinch, and dense plasma focus (DPF) have been invented and used for various purposes over the decades of development of the pulsed power family [1–5]. DPFs have become efficient sources of fusion neutrons [5, 6]. Such devices are characterized by a long current rise time (on the order of units of μ s), and their operating regime is strongly influenced by the quality of the insulator and the dynamics of the current sheath formation process [7–9]. A notable phenomenon observed in mega-ampere-class DPFs is the saturation of neutron yield effects [10–12]. This is a key factor contributing to the current preference for smaller devices powered by kJ- or sub-kJ-level energy storage systems [13–15].

X-pinchs, which were introduced in the 1980s, aim to achieve an intense source of x-rays [16–19]. The configuration is uncomplicated and consists of two electrodes shunted by two or more crossing wires that touch in a single point, thus giving the configuration the appearance of the letter ‘X’. The primary application of the X-pinch is the generation of a point-like source for x-ray radiography and other applications [17–19]. Achieving this objective places a considerable demand on the generator with respect to the current rise time. Shelkovenko *et al* quantified this demand to be [20]

$$\left(\frac{di}{dt}\right)_{\text{average}} > 1 \text{ kA ns}^{-1}. \quad (1)$$

As with other configurations, the X-pinch presents several drawbacks. Setting aside the complexity of thin fiber fabrication and the challenges of precise alignment, the primary complication with megaampere generators arises when the fibers become overloaded. While such a current overload can be mitigated by adding extra fibers, this brings a new challenge: ensuring a single, well-defined contact

point among multiple fibers. This is not always reliably achieved through the simple twisting or rotation of the wire array holder. To overcome these limitations, hybrid X-pinch configurations have been increasingly employed. This design incorporates two conical electrodes forming the characteristic 'X' shape, between which a thin wire or fiber of material similar to that used in conventional X-pinch is placed [21, 22]. Although primarily used as point sources, hybrid X-pinchs have also been utilized for neutron and ion generation, for example, by employing deuterated CD_2 fibers [23, 24].

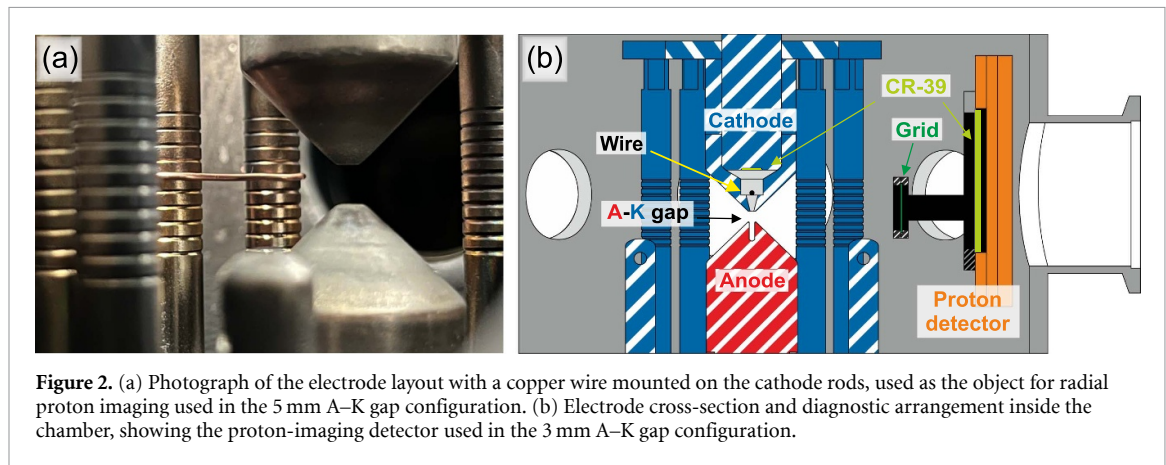
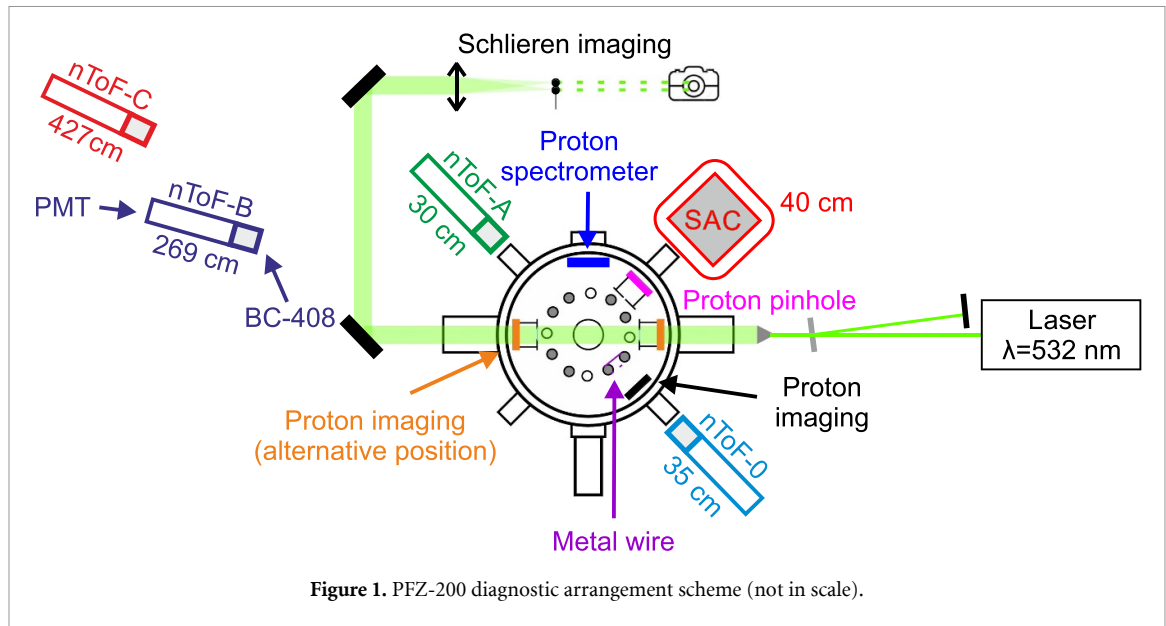
In this paper, we focus on the hybrid X-pinch with a modification in which a single wire is replaced by deuterium gas filling, creating a gaseous hybrid X-pinch driven by a DPF. Such a configuration neglects the need to change the wire after each shot, thereby reducing the time between shots by keeping the experimental chamber closed. Another advantage of the gaseous hybrid X-pinch is that the mass density can be optimized by adjusting the gas pressure rather than changing the wire material at different densities. The challenge of experimental load alignment with the X-pinch remains, as wire alignment is replaced by precise conical electrode positioning. The experiments described in this paper were conducted on a PFZ-200 plasma focus at the Czech Technical University in Prague [25]. The generator consists of four capacitors switching through four spark gaps with no pulse shape modification in the transmission line. The current peak rise time is less than $2\ \mu\text{s}$. For the peak current of 200 kA, the average current derivative is about $0.1\ \text{kA ns}^{-1}$, which is much lower than the condition (1) for reaching a bright spot with ns duration. Nevertheless, such a long rise time includes several phases of the DPF device, such as the breakdown at the insulator surface and axial acceleration of the current sheath. However, the most critical phase for the hybrid X-pinch is when the current increases between the conical electrodes during the radial compression phase. In this stage, the current reconnects from flowing between the anode and the radial cathode rods to flowing between the anode cone and the axial cathode cone, where the current derivative is much higher locally in the narrow region between the cone tips.

This study aims to develop and characterize a proton point-like source that can be used as a backlighter for proton imaging and deflectometry, both in plasma research and other interdisciplinary applications [26, 27]. The paper is organized as follows: in section 2, the PFZ-200 device and diagnostics are described; section 3 presents the hybrid X-pinch plasma visualization; section 4 briefly summarizes the neutron production during the experiments; section 5 is the main part of this paper, where the proton source analysis results are presented and discussed, including proton imaging, proton pinhole and proton spectra measurement.

2. Experimental setup and apparatus

The experiment was performed on a 3 kJ plasma focus PFZ-200 with a discharge current maximum above 200 kA and a $1.6\ \mu\text{s}$ rise time. The electrode system was designed as a Mather type, where the anode rod was 150 mm long, with a diameter of 25 mm, and ended with a conical tip [25]. The anode was coaxially surrounded by a cathode composed of 12 stainless steel rods with a 8 mm diameter. Four of the rods were shortened to 150 mm (from the standard of 200 mm) to allow diagnostic access in the radial direction. To establish the hybrid X-pinch, the cathode was extended by the axial cone, positioned axially and above the anode (see figure 2(b)). The axial cathode was mounted on a holding platform supported by the cathode rods. The anode and cathode cones are made of the tungsten-copper alloy with a ratio of 80 % W and 20 % Cu. Both anode and cathode cones are tapered at 45 degrees to the tip diameter of 2 or 4 mm depending on the axial anode–cathode (A–K) gap settings. There is a hole at the tip of each cone, forming a hollow tapered shape. The axial A–K gap between the cones was set to 3 and 5 mm, and the A–K gap in the radial direction (between the anode and cathode rods) was 13 mm. The electrode system was installed inside a vacuum chamber, and all measurements were performed in deuterium gas at a pressure of 360 Pa. This value was chosen on the basis of prior pressure-optimization studies conducted with different anode tip configurations, which identified 360 Pa as the pressure that maximizes the neutron yield [25].

Several ion detectors were employed in different configurations to characterize the DD-fusion proton source (see figure 1 and 2). Radial and axial proton imaging of a testing object (wire or plastic grid) is used to analyze the proton source. The goal was to measure the source dimensions based on shadow properties. For axial imaging, it is necessary to machine the cathode cone to provide an accessible location for CR-39, as shown in figure 2. We also measured the proton-energy spectra using an absorption spectrometer, which consisted of CR-39 shielded by several aluminum foil stripes of different thicknesses. Owing to the lack of space in the axial direction and lower durability of the thin aluminum foils, the spectrometer was placed in the radial direction. In addition, we used the CR-39 pinhole detector with a pinhole diameter of 0.9 mm in the radial direction. Considering the typical proton yields, the signal from this detector accumulated throughout the experimental campaign. All the CR-39 plates were etched



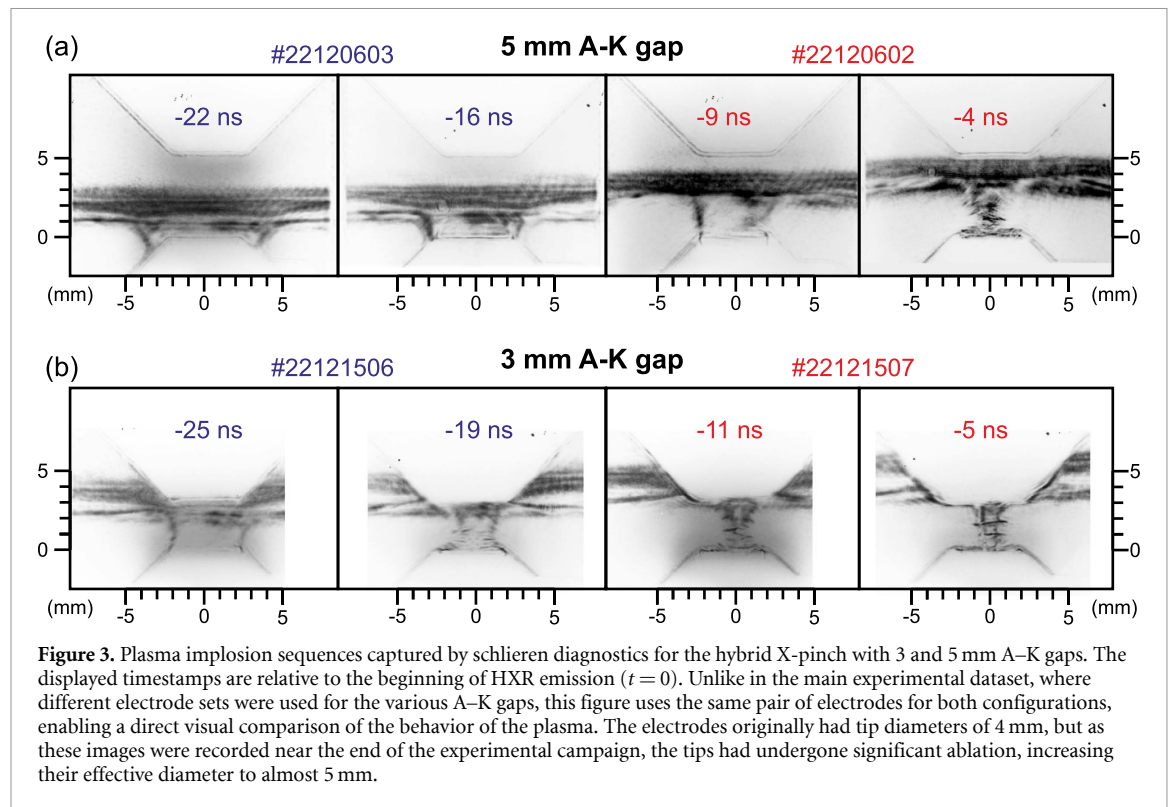
after exposure to 6 M NaOH for 6 h at 80 °C. CR-39 analysis was conducted using an Epson Perfection V850 Pro optical scanner and a Leica DVM6 digital microscope.

Neutron diagnostics contain a silver activation counter (SAC) based on the Los Alamos model [28, 29] supported by scintillation neutron-time-of-flight (nToF) detectors with a photomultiplier tube [30]. All the detectors were placed in the radial direction at different distances with respect to the discharge axis (see figure 2(b)). The nToF detectors consisted of a BC-408 scintillator 2 inches in diameter and 5 cm in length. The scintillator was optically coupled to the photomultiplier Hamamatsu H1949-51, which was supplied with a high-voltage in the range of 1.0–1.5 kV. During our experiments, we used nToF detectors for neutron and hard x-ray detections. The neutron production time was registered by closer detectors at 30 and 35 cm from the pinch axis. The further detectors at 2.5 and 4.3 m were primarily used for neutron energy and HXR evaluation.

The discharge current and its derivative were measured using a Rogowski coil placed in a plasma focus collector. The plasma implosion was visualized by the schlieren technique using the 532 nm Nd:YAG laser with (2.0 ± 0.5) ns pulse duration and maximal energy of 87 mJ.

3. Plasma visualization

Experimental research was conducted on a PFZ-200 pulsed power device, where the hybrid X-pinch configuration was obtained by deuterium gas implosion between the two conical electrodes. During our experiments, we used laser schlieren diagnostics to observe the behavior of the plasma in the A-K gap and to check the quality of the plasma compression.



The captured plasma implosion images are shown in figure 3. The figure consists of two sets of images mapping the temporal pinch evolution with 5 and 3 mm gaps between the conical electrodes, where each set consists of data from two different discharges displayed in chronological order.

If we compare the hybrid X-pinch plasma column driven by the DPF with typical experiments performed on such a device [25], we can see that the pinch column is significantly shorter. When the DPF used a cylindrical anode tip, the typical column length increased to 10 mm. The column length was reduced to approximately 6 mm by using a conical anode without an axial cathode. During the hybrid X-pinch experiments with a 5 mm A-K gap, the length of the pinching plasma is limited by the opposite cone; however, the plasma implodes and disrupts before reaching the cathode cone. If we look at the last time frame of figure 3(a), we can see that there is only a 3 mm pinch with approximately 2 mm thick current sheath above. The sheath moves upwards along the z -axis, and the pinch could be elongated when the sheath passes the cathode edge. However, we have never registered such a pinch stretched within the full A-K gap. On the basis of this finding, we assumed that there was an opportunity to set the electrodes closer.

The results from the shots with the 3 mm A-K gap shown in figure 3(b) clearly show that the edge of the current sheath reaches the cathode at earlier times and subsequently implodes radially. As the plasma was compressed, the current sheath continued to move axially while slightly expanding along the conical cathode. Pinch deformations were observed in few shots, where the plasma column appeared to be deflected or partially ejected prior to the final compression. However, these deformations are rare.

4. Neutron production

We expect a tight correlation between neutron and proton emissions based on the cross-section of the DD fusion reactions [31]. Therefore, we measured the neutron yields by SAC to estimate the proton yields immediately after the shot. We conducted two experimental campaigns with 3 and 5 mm A-K gap. Approximately 60 discharges for each configuration were performed. Due to the need to exchange the proton detector, the chamber had to be opened after each shot that produced a neutron yield exceeding 2×10^7 was necessary. Based on the plasma focus device behavior, which is significantly affected, by oxidation of the insulator's metallic contamination, performing a shot with sufficient neutron yield within the first shot after chamber closing was challenging [25]. Neutrons are usually registered after the second, or third, shot. Therefore, this procedure reduced the number of shots suitable for data evaluation to 27 with a 5 mm A-K gap and 22 with a 3 mm A-K gap.

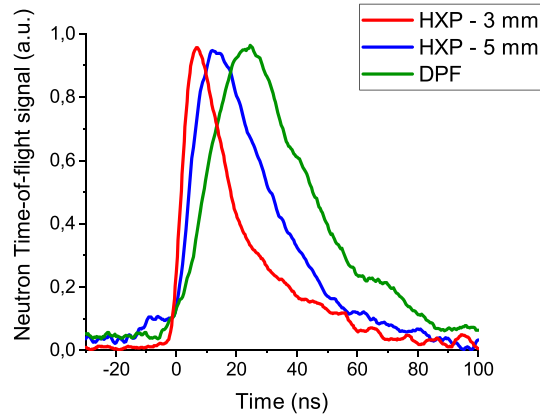


Figure 4. Neutron production times measured by neutron time-of-flight detectors during the experiments with 3 and 5 mm A–K gap hybrid X-pinch and unmodified plasma focus configurations. The detectors were positioned at the distance of 30–35 cm in the radial direction.

Table 1. Neutron production times (full width at half maximum) measured by neutron Time-of-Flight detectors during the experiments with 3 and 5 mm A–K gap hybrid X-pinch and unmodified dense plasma focus configurations.

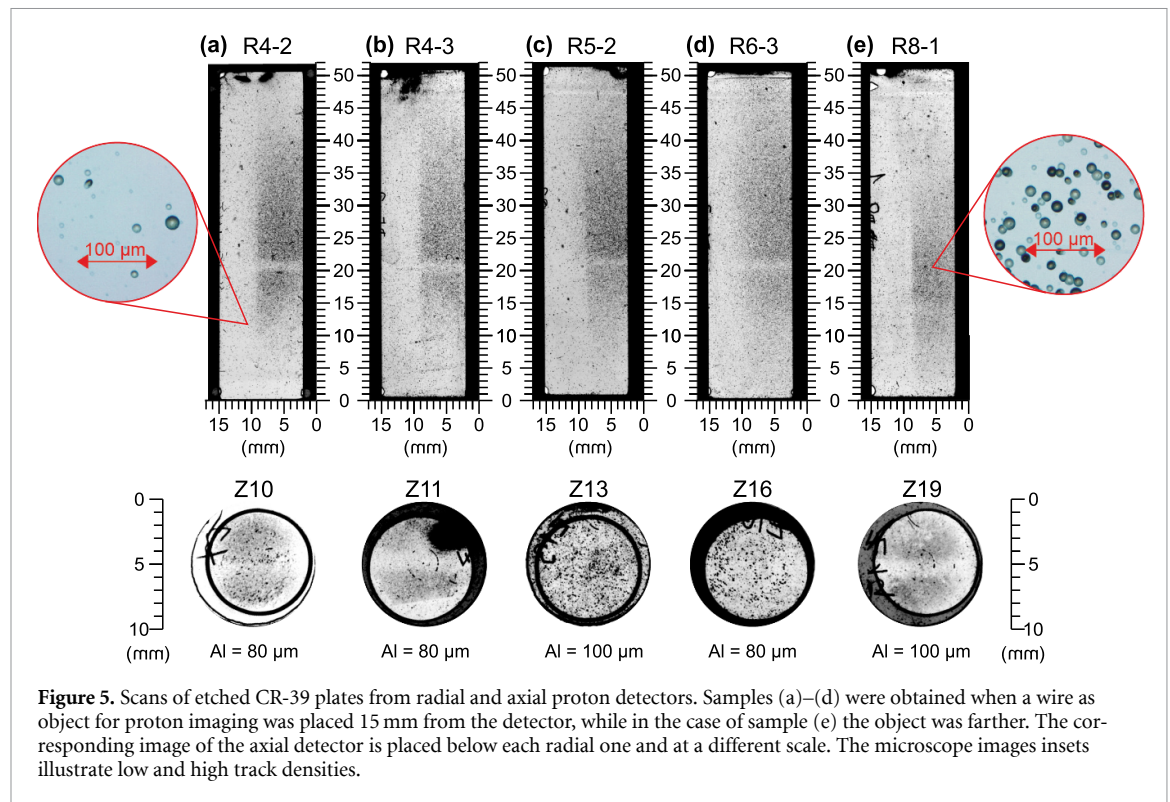
Configuration	Neutron production time FWHM (ns)
Hybrid X-pinch with 3 mm gap	20 ± 7
Hybrid X-pinch with 5 mm gap	27 ± 8
Dense plasma focus	38 ± 9

The typical neutron yield during the experiments varied between 10^7 and 10^8 neutrons per shot. Compared to the common operation of the PFZ-200 without the hybrid X-pinch modification we registered a lower neutron yield by 40%–50%. The maximum neutron yield produced by the hybrid X-pinch was 10^8 neutrons/shot. However, after neglecting each 1st shot in the series, the average yield was approximately 6×10^7 neutrons/shot.

The neutron production times were measured using a scintillation time-resolved detector. Two detectors are employed to achieve better data accuracy. Owing to the sensitivity of the BC-408 scintillator to x-rays, the detectors were shielded by 5 cm thick lead bricks to prevent saturation of the photomultiplier. The production time was evaluated as the full width at half maximum (FWHM) of the recorded neutron peak. The following experimental loads were tested: a hybrid X-pinch with 3 and 5 mm A–K gap and an unmodified plasma focus without the opposite conical electrode. Representative signals for each configuration are shown in figure 4. Through statistical analysis, we evaluated the mean values of the production times and their standard deviations. Neutron production time was reduced by limiting the space above the anode with the cathode positioned opposite. According to table 1, the longest time measured with the unmodified plasma focus was (38 ± 9) ns. When the axial cathode cone was placed 5 mm from the anode resulted in a time of (26 ± 6) ns. Moving the cathode closer to the 3 mm A–K gap the production time is reduced to (20 ± 7) ns.

5. Proton source characterization

The main objective of our research is to characterize the proton source that originates in the beam-target DD fusion reactions. For this purpose we performed proton imaging using a set of CR-39 track detectors with aluminum foils to absorb low-energy protons and ions. Based on the calculations in the SRIM software [32], we chose the 70 μm thick aluminum foils to filter protons with an energy below 2.75 MeV. An aluminum layer of such a thickness also absorbs the deuterons with an energy below 3.5 MeV. As described in section 2, we placed the detectors in the axial and the radial directions. The CR-39 detector was positioned close to the proton source (1.5 cm) in the axial direction. Owing to the hybrid X-pinch configuration, the detector was housed inside the cathode cone. Despite the 1–2 mm hole diameter in the electrode, the aluminum foil was highly exposed to the discharge, which led to its degradation. For this reason, we increased the filter thickness to 80 μm and subsequently to 100 μm to prevent damage to the detector. Such a modification increases the cut-off energy from 2.75 MeV to 3 MeV and 3.5 MeV, respectively. However, we expect the filter to undergo degradation and a reduction of its effective thickness to (50–70) μm depending on the number of experimental shots.



For further analysis of CR-39 detectors, achieving a track density of approximately 10^5 cm^{-2} is advantageous for preventing overlapping proton traces [33]. Given that the farthest detector was located 5 cm radially from the pinch axis, an accumulated proton yield of approximately 3×10^7 was required. The main goal was to characterize the individual shots and minimize the signal spread and shadow overlapping caused by, e.g. shot-to-shot fluctuations of the source position or source size. The conditions in our device do not allow the production of a sufficient number of fusion protons in the first shot after closing the vacuum chamber. However, in the experiments with both 3 mm and 5 mm A–K gap configurations, the second shots after chamber closure produced sufficient numbers of protons and neutrons, whereas the yield from the first shots was at least an order of magnitude lower or even negligible. Based on this observation, we assumed that the proton source could be successfully characterized in a single shot. In some cases we accumulated protons to obtain a higher contrast in our detectors. The following paragraphs describe the proton emission for two different A–K gaps between the axial conical electrodes.

5.1. 5 mm A–K gap

The first attempt to limit the volume of the potential proton emission was to mount an opposite conical electrode in the z-axis at the distance of 5 mm from the conical anode tip to create a 5 mm A–K gap. According to schlieren diagnostics, the original maximal pinch length (without the opposite cone) varied between 5 and 6 mm with a conical anode and up to 10 mm with a cylindrical anode tip [25]. When the opposite cone was applied, the maximal pinch length was reduced to 3 or 4 mm (see figure 3). Nevertheless, the maximum discharge current and its waveform recorded by the Rogowski coil were not affected by such a pinch reduction. On the other hand we observed lower neutron yields, on average. Generally the typical yields without the opposite cone were approximately $(1.0 - 1.5) \times 10^8$ neutrons/shot. After modification into a hybrid X-pinch the yields were reduced to the range of $(2 - 9) \times 10^7$ neutrons/shot.

During the experiments with a 5 mm A–K gap, we used a proton imaging in the axial and radial directions with a copper wire with a diameter of 1 mm. In addition, we used a proton pinhole detector where the signal accumulated throughout all shots of the experimental campaign.

The proton imaging results from the radial and axial directions are shown in figure 5. The radial results are presented in the upper part of figure 5 and the complementary axial detectors are shown below (circular detectors). We collected over twenty CR-39 plates from radial imaging. However, for the purpose of this study, we selected only a reliable subset that effectively represented the observed proton source. First, CR-39s were analyzed using an optical scanner after chemical etching. Based on these scans, we evaluated the gray value plot profiles of each detector. Such a scanning technique does

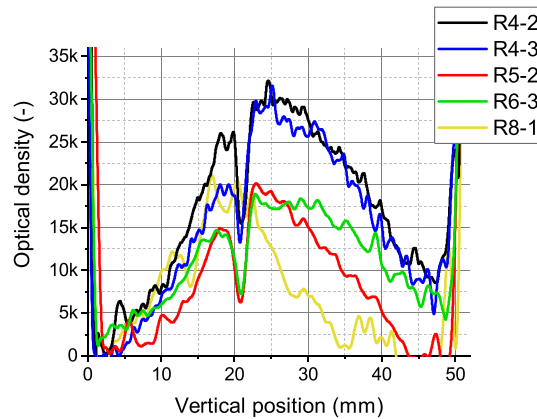


Figure 6. Gray value profiles of proton radial detectors displayed in figure 5.

not provide information regarding individual tracks and their diameters. Briefly, the gray value profiles explicitly do not show if a specified darker area has a higher track density of smaller tracks or a lower density of larger tracks. Therefore, we subsequently analyzed CR-39s using a digital microscope then we applied our script in MATLAB for track recognition. The main reason for analysis by both techniques is the varying track diameter, which depends on several factors including the type of particle, its energy, or the etching properties e.g. etching time, etching temperature, the kind of the etching solution, and its concentration [34–37]. Our CR-39s were etched in 6 M NaOH for 6 hours at 80 °C, which represents the same etching conditions applied throughout this study and ensures consistency in the comparison of track properties.

From the proton images displayed in figure 5, one can recognize the lack of a significant signal on the left-hand side of the radial detectors. Such a signal decrease is caused by the cathode rods that screen the detector. In all samples, excluding 5(d), the cathode rod shadows have a sharp edge. In the example shown in figure 5(d), the edge is much smoother, which may be caused by the larger size of the proton source in the horizontal orientation.

The primary objective of proton imaging is to investigate the proton source and evaluate its dimensions. The small horizontal shadows from the metal wire in CR-39 are relatively well recognizable in figures 5(a)–(d) in their lower parts. In the case of sample (e), the shadow appears faint because the wire was shifted 15 mm from the source and farther from the detector, and due to the proton source properties. Detector 5(e) was exposed to four shots, and the highest neutron yield of 9×10^7 was obtained in the fourth shot after closing the chamber. The previous shots had no yield, or less than 1×10^7 neutrons/shot.

Scanning with an optical scanner provides a fast macroscopic view of the entire CR-39 detector under uniform backlighting conditions. However, track recognition using digital microscopy is a more valuable approach for analyzing proton sources, as it enables the identification of every individual track. In contrast, obtaining a macroscopic view with microscopy requires stitching together multiple images, which complicates post processing owing to varying backlighting conditions or loss of focus caused by the corrugated surface of the detector. The gray value profiles are plotted in figure 6, and track density profiles obtained via digital microscopy are presented in figure 7.

Both graphs (figures 6 and 7) demonstrate that the maxima of the measured signals were in the middle of the detector at approximately 25 mm. Both plots also agreed that the profiles had reached the background level at their ends. The projections of the wire shadows were recognizable between 19–22 mm. Only in sample R8-1 is the maximum shifted to nearly 20 mm because the detector was asymmetrically rotated. Despite the fact that the wire shadow is well recognizable, based on both profile plots, the signal behind the wire is not reduced to the background level, as on the sides.

The experimental setup of radial proton imaging is shown in figure 8, which illustrates three different source sizes that produce distinct signal profiles behind the 1 mm thick wire. As indicated in figure 8, the critical source diameter was approximately 3.3 mm. A smaller source results to an umbra with a drop of the signal to the background level, whereas a larger source generates only a penumbral projection. It is important to note that this simplify interpretation assumes a homogeneous source with a sharp boundary.

For a more realistic scenario involving a non-uniform proton source with a Gaussian spatial distribution, we conducted Monte Carlo simulations. The resulting signal profile is plotted in figure 9,

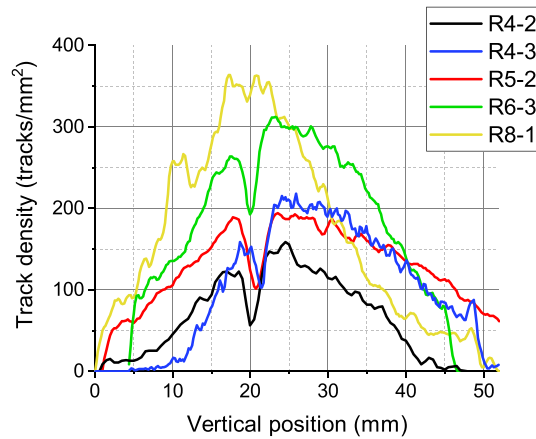


Figure 7. Linear proton density profiles in CR-39 detectors shown in figure 5.

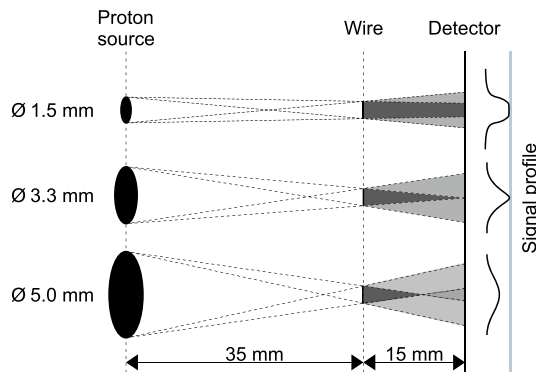


Figure 8. Illustrative scheme of proton source size analysis in respect to real experimental scale.

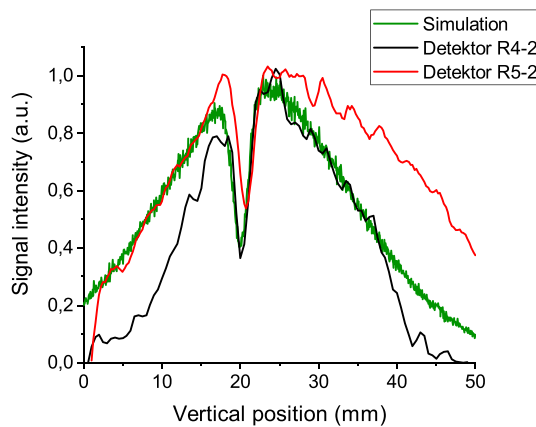


Figure 9. The comparison of measured proton signals by the radial CR-39 detectors and simulation of proton source with Gaussian profile with FWHM = 5 mm.

where it is compared with the experimental penumbra profiles obtained using the two detectors shown in figure 7. The simulation was performed for a Gaussian source with a FWHM of 5 mm. These findings suggest that a wire with such small dimensions is inadequate for a precise source analysis. Consequently, the object was redesigned for future experiments, as described in section 5.2.

In addition to the radial perspective, it is also important to characterize the source size along the axial direction. As shown in figure 5, axial CR-39 registered an intermittent signal, which was potentially influenced by a relatively small hole in the cathode cone and variations in the source position. Figure 5 shows the five detectors. Three of these showed measurable signals alongside the shadow of the wire.

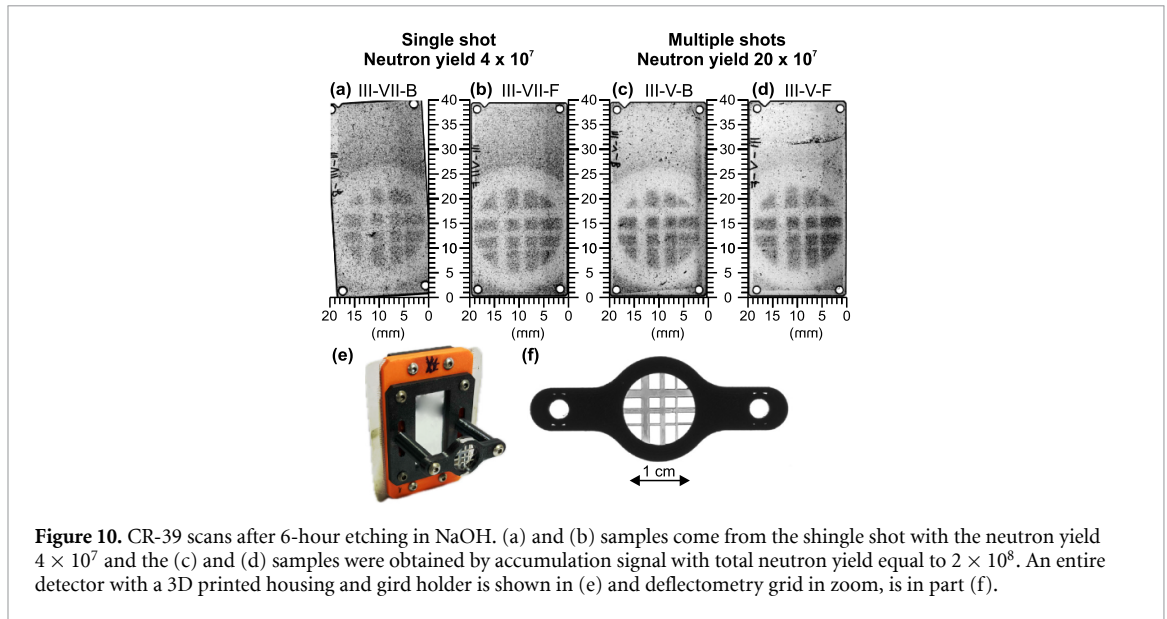


Figure 10. CR-39 scans after 6-hour etching in NaOH. (a) and (b) samples come from the shingle shot with the neutron yield 4×10^7 and the (c) and (d) samples were obtained by accumulation signal with total neutron yield equal to 2×10^8 . An entire detector with a 3D printed housing and grid holder is shown in (e) and deflectometry grid in zoom, is in part (f).

At least one of these signals (sample Z11) exhibited a profile recognizable as an umbra with penumbras, enabling a valid evaluation of the source size. This analysis indicated a source size of approximately 1.5 mm.

5.2. 3 mm A–K gap

After a series of shots with a 5 mm A–K gap, we tested a new pair of conical electrodes and placed them closer to each other with a 3 mm gap. Because of the width of the cone tips (which was previously approximately 3 mm), we used electrodes with the same angle but a smaller tip diameter of 2 mm. During the experiments with a 3 mm gap, we improved the object for proton imaging. The metal wire was replaced by a deflectometry grid made of CR-39 (see figure 10(f) with horizontally and vertically oriented bars of three different widths (1.3, 0.8, and 0.35 mm). The thickness of the CR-39 grid was (0.50 ± 0.06) mm, which is sufficient to stop 3 MeV protons. Such a grid was directly attached to the CR-39 housing to minimize the uncertainty in the grid-detector mutual position in the distance of 20 mm. The distance between source and the grid was 40 mm.

As shown in figure 1, we placed two detectors facing each other for proton imaging in the radial direction. Selected results from proton imaging with a 3 mm A–K gap are displayed in figure 10, where optical scans of etched CR-39 plates are presented. In figures 10(a) and (b) the proton images are obtained from a single shot with a neutron yield of 4×10^7 , whereas signals in figures 10(c) and (d) are accumulated from multiple shots with a total neutron yield of 2×10^8 . All samples showed relatively sharp grid shadows. The most distinct shadow (umbra) is observed behind the thickest bar, where a full shadow forms, allowing us to clearly distinguish between umbra and penumbra. As shown in figure 10, the horizontal and vertical middle bars were also projected, although the signal decreased to approximately the background level. The signal minima behind the middle bars were narrower than those behind the thickest bars. Despite the lower attenuation, the signal still decreases close to the background, when practically all the penumbra is registered. Based on these findings we could, without other data evaluation, estimate that the analyzed proton source is of a smaller size compared to the results from section 5.1, where the signal behind the larger object (by 0.2 mm thicker) was more defocused. Behind the thinnest bars, the shadow was still observable. However, it is a weak signal with a minimal possibility for further source analysis.

Similar to section 5.1, we analyzed the CR-39 plates using optical scanning and digital microscopy. Subsequently, the gray values and proton density profiles were plotted and analyzed as shown in figures 11 and 12 respectively. The summarized results are presented in table 2, where the source dimensions were calculated from the penumbra widths, as marked in figure 11. Based on the obtained profiles, we only assumed shadows behind the large and middle bars. Partial shadows behind thinnest bars were insufficient for further full-feature evaluations. The values in table 2 are sorted by detector names, bar thicknesses, and orientation of the measured source dimension, e.g. vertical (ver) refers to a horizontally oriented bar creating an umbra, which was measured along the vertical axis corresponding to the vertical dimension of the proton source. The dimensions were evaluated from both left-hand (LH) and right-hand (RH) penumbras.

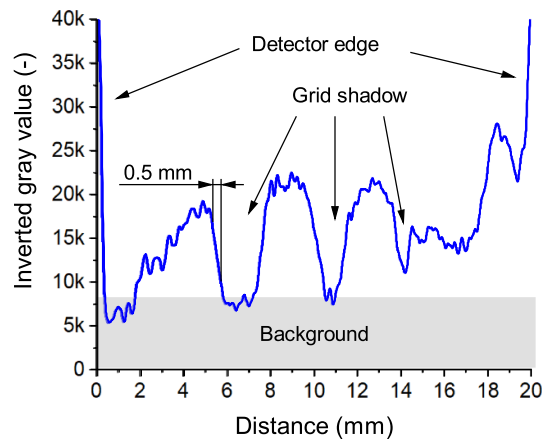


Figure 11. A gray value profile of a CR-39 plate in figure 10(c).

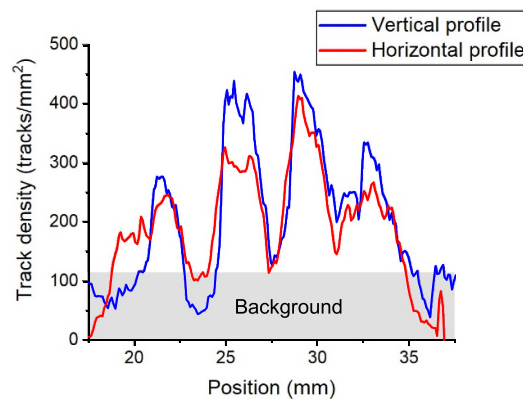


Figure 12. A track density profiles of a CR-39 plate from figure 10(c).

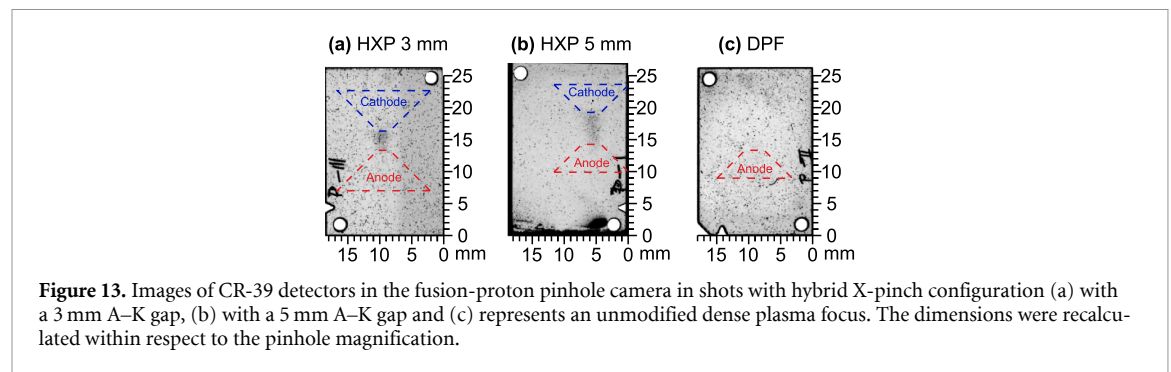
Based on the values displayed in table 2, one can conclude that the dimensions of the proton source generated by the hybrid X-pinch with a 3 mm A–K gap fluctuate between 0.8–2.0 mm. If we compare both measuring techniques, scanning, and microscopy, we see that they are consistent, at least within their uncertainties. The values obtained from the single-shot experiment (III–V-B/F) indicate to the proton source height of (1.3 ± 0.2) mm and width of (1.4 ± 0.3) mm. These findings lead to the conclusion that the source cross-section is circular. In the case of the multiple-shot results (III–VII-B/F), the source width and height is equal to (1.1 ± 0.1) mm from scanning technique. Using the microscope technique, we obtain the same average width value; however, the height is measured to be larger (1.4 ± 0.3) mm, though still consistent within its larger uncertainty.

5.3. Proton pinhole results

As mentioned in section 2, a proton pinhole detector was placed inside the experimental vacuum chamber in the radial direction to accumulate fusion protons throughout each experimental campaign. The pinhole diameter was $900 \mu\text{m}$, and the CR-39 plates were shielded with $70 \mu\text{m}$ thick aluminum foils. This means that we registered protons with energies above 2.75 MeV. A comparison of all electrode configurations is shown in figure 13. Such highly localized signals are clearly observed in figures 13(a) and (b). However, as shown in figure 13(c) the signal is spread within all detector plate. Based on knowledge of the source-pinhole and pinhole-detector distance ratio, we can calculate the projection magnification as $\frac{27 \text{ mm}}{37 \text{ mm}} = 0.73$. By analyzing the signal in figure 13 we observed that protons were emitted in the entire axial A–K gap during the campaigns. However, the highest signal was localized closer to the cathode, and in the case of the 3 mm A–K gap, it covers approximately 1.5–2.0 mm. In the 5 mm gap, the signal is more uniformly distributed, however the lower part of the source detected a slightly weaker signal. These findings confirm the results of proton imaging.

Table 2. Measured proton source dimensions and the comparison of analyses using the gray profiles and particle densities from microscope. The source dimensions were calculated based on the penumbra widths behind the large (1.3 mm) and middle (0.8 mm) bars. The term ‘orientation’ refers to the measured dimensions, not the bars themselves. For example, ‘ver’ refers to a bar that is oriented horizontally and to the vertical dimension. The dimensions were evaluated from left-hand (LH) and right-hand (RH) penumbras.

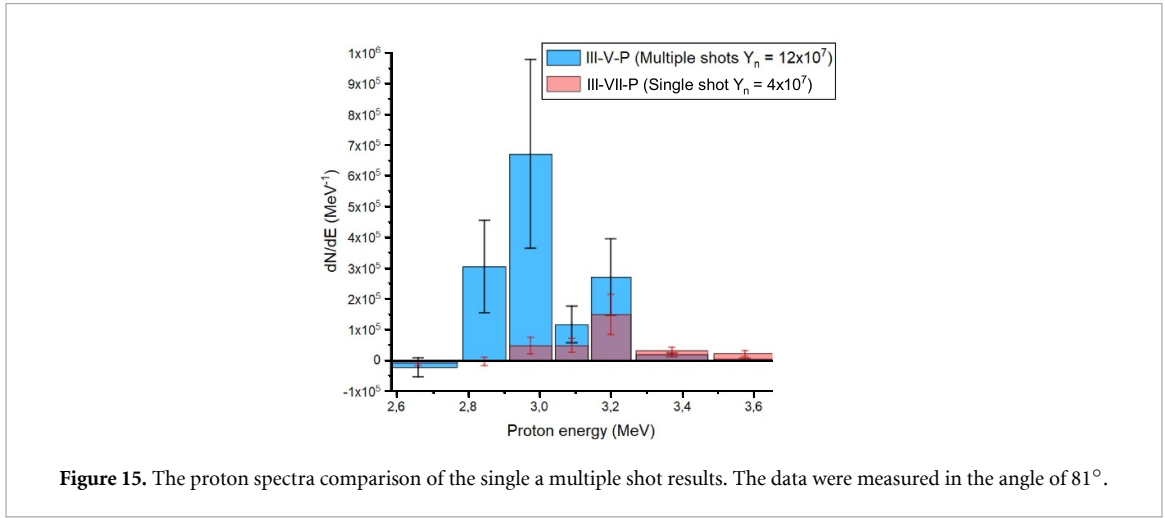
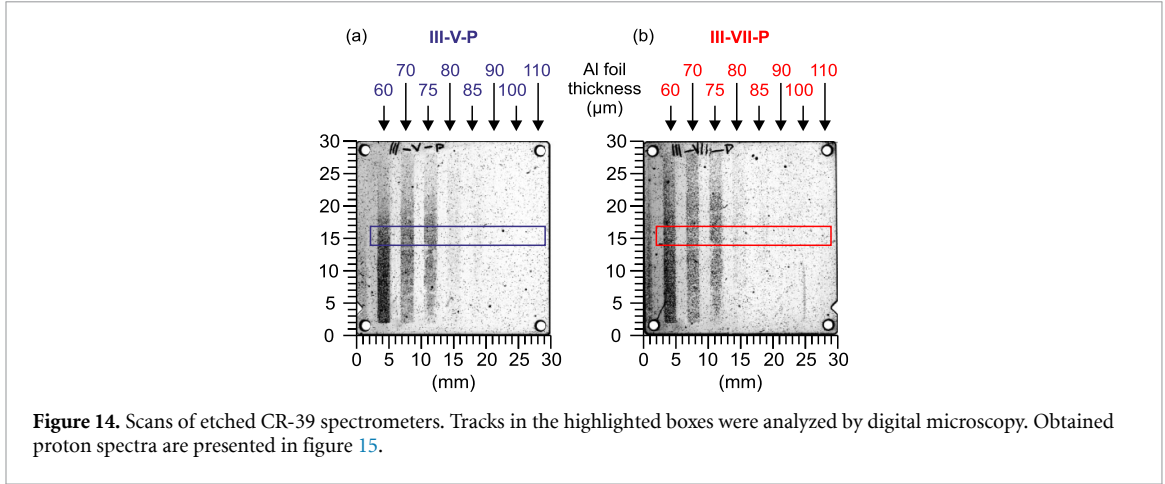
Detector			Source size (mm)			
Name	Thick (mm)	Orientation	Scanner		Microscope	
			LH edge	RH edge	LH edge	RH edge
III–V-B	1.3	ver	1.2	1.0	1.6	1.2
		hor	1.0	1.6	1.8	1.4
	0.8	ver	1.4	1.6	1.2	1.6
		hor	1.6	1.2	1.8	1.2
III–V-F	1.3	ver	1.6	1.6	1.4	1.4
		hor	0.8	1.6	1.0	2.0
	0.8	ver	1.0	1.2	1.2	1.0
		hor	1.4	1.2	1.2	1.4
III–V-B/F Summary		ver	1.3 ± 0.2		1.3 ± 0.2	
		hor	1.3 ± 0.3		1.5 ± 0.3	
III–VII-B	1.3	ver	1.4	1.2	1.2	1.2
		hor	1.2	1.2	1.0	1.4
	0.8	ver	1.0	1.2	1.6	1.6
		hor	1.2	1.0	1.0	1.2
III–VII-F	1.3	ver	1.0	1.2	1.6	1.6
		hor	1.2	1.0	0.8	1.0
	0.8	ver	1.0	1.0	1.2	0.8
		hor	1.0	0.8	0.8	0.8
III–VII-B/F Summary		ver	1.1 ± 0.1		1.4 ± 0.3	
		hor	1.1 ± 0.1		1.1 ± 0.2	



5.4. Proton spectra

Using the DD fusion proton spectrometer with CR-39 and aluminum foils of different thicknesses, we measured the proton spectra from the hybrid X-pinch with a 3 mm A–K gap. The detector contains foils with thicknesses of (60; 70; 75; 80; 85; 90; 100 and 110) μm corresponding to the proton energy range (2.54–3.67) MeV. The detector was placed in the radial direction, but owing the detector dimensions and experimental arrangement, the presented spectra (see figure 15) were measured at an angle of 81° instead of 90° .

The CR-39 detector is sensitive to a wide range of particles. In this context, it is important to clarify whether the measured signal originates from protons or deuterons. We attribute the measured signal to protons rather than deuterons based on the following arguments. The spectrometer is sensitive to protons within 2.54–3.67 MeV and to deuterons within 3.25–4.75 MeV. However, it is unlikely that deuterons would be accelerated to such energies in this type of device. In our configuration, deuterons in this range would correspond to proton energies above 4.25 MeV, which have never been observed. The results point to a relatively monoenergetic proton spectrum centered around 3 MeV, whereas accelerated deuterons would be expected to produce a monotonically decreasing spectral profile. The signal in figure 14 is fairly isotropic and vanishes smoothly in both directions, which is consistent with fusion protons being



emitted into 4π . If deuterons were detected, the signal would be expected to exhibit pronounced anisotropy, as reported in other experiments (e.g. those performed on MAIZE at UM)[23], due to the formation of collimated deuteron beams. Furthermore, the proton signal scales with the neutron yield, which provides additional support for the conclusion that the detected particles are fusion protons rather than accelerated deuterons.

Figure 15 shows a comparison of the measured proton spectra from the single- and multiple-shot series. As shown in figure 15, the evaluated proton yield from the single shot was 2×10^7 , and the accumulated proton yield from multiple shots was 9×10^7 protons. These values are approximately two times lower than the neutron yields measured using SAC. This relatively large discrepancy could be caused by the proton and neutron anisotropy. Although both the detectors were placed in the radial direction, they were located at different positions. Moreover, the measurement may have been affected by scattered neutrons, leading to an overestimation of the total neutron yield by the SAC. This possibility should be considered when interpreting the SAC neutron yield data.

Comparing both spectra from single and multiple shots, one could observe the dominant energy of 3.2 MeV in the single-shot data. In the multiple shots, we observe 3.2 MeV protons too. However, the major peak shifted to ≤ 3 MeV. In the case of the static target the fusion proton energy $E_p(E_d, \theta)$ depends on the incoming deuteron energy and the angle between the incoming deuteron and the outgoing proton, which is described by the following equation:

$$E_p(E_d, \theta) = \frac{E_d m_d m_p}{(m_T + m_p)^2} \cdot \left(\cos \theta + \sqrt{\frac{m_T E_d (m_p + m_T - m_d) + Q m_T (m_T + m_p)}{E_d m_d m_p} - \sin^2 \theta} \right)^2, \quad (2)$$

where m_d , m_p , and m_T are the incoming deuteron, outgoing proton and triton masses, respectively. The Q represents the reaction energy, equal to 4.03 MeV. By solving equation (2) for a proton energy of

3.2 MeV and angle $\theta = 81^\circ$, we found that protons were produced from collisions of deuterons accelerated to energy reaching 300 keV. According to figure 15, the maximum measured proton energy is 3.6 MeV in both single and multiple shots, indicating deuteron energies of up to 1.3 MeV for the same collision angle.

In figure 15, it is clear that during the multiple-shot experiment, a higher number of lower-energy protons was produced than in a single shot. However, an unpredictably low energy of ~ 2.85 MeV was measured. The expected proton energy for such a collision angle should not be less than 3 MeV. In such multiple-shot experiments, it is possible that some shots had off-axis proton sources, which could have affected the collision. This opens up the view of the angle variation. If we assume 300 keV deuterons, the 2.85 MeV protons should be emitted at an angle of 110° .

6. Conclusion

We successfully tested and characterized a hybrid X-pinch on a 3 kJ DPF device by modifying the electrode system to include two axially aligned conical electrodes. Unlike conventional loads, which typically employ metal wires or fibers, this configuration utilizes deuterium gas filling. Experiments were conducted with A–K gaps of 5 and 3 mm, with the primary objective of developing a point-like proton source and evaluating its suitability for proton imaging.

Experiments with a 5 mm A–K gap confirmed the generation of an intense proton source in the region between the conical electrodes. However, proton imaging and pinhole camera measurements indicated source scale of up to 5 mm, resulting in relatively blurred images of a 1 mm metal wire.

In contrast, experiments with the 3 mm A–K gap yielded improved results. Proton imaging of a plastic grid with bars of different thicknesses confirmed the generation of a circular proton source with a diameter between 1.1 and 1.5 mm and a yield of the order of 10^7 . The proton energy spectra measured in the radial direction using CR-39 detectors and aluminum filters of different thicknesses revealed maximum proton energies of up to 3.6 MeV, corresponding to deuteron energies approximately up to 1.3 MeV.

Data availability statement

The data that support the findings of this study are openly available at the following URL: <https://drive.google.com/drive/folders/1vzcIwpiTt6-IZUKPJtZDvqdVezorqcJo?usp=sharing> [38].

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